

Business Requirements Document (BRD)

Olist E-Commerce Analysis

Project Type: E-Commerce Business Intelligence & Analytics

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Domain: E-Commerce

Primary Tools: SQL (MySQL), Power BI, DAX, Power Query

1. Document Information

Field	Details
Document Name	Olist E-Commerce Analysis – Business Requirements Document
Project Name	Olist E-Commerce Analysis
Document Type	Business Requirements Document (BRD)
Version	v1.0
Author	Sejal Sayam
Date	August 2026
Business Domain	E-Commerce
Reporting Solution	SQL + Power BI
Dataset	Brazilian E-Commerce Public Dataset by Olist

1.1 Document Purpose

This document defines the business requirements for an e-commerce analytics and reporting solution designed to provide stakeholders with centralized visibility into sales, customers, products, sellers, payments, reviews, and delivery performance.

The requirements are based on the business problem and analytical objectives identified in the Olist E-Commerce Analysis project.

1.2 Document Objective

The objective of this BRD is to clearly document:

- The business problem being addressed.
- The objectives of the analytics solution.
- The scope of the project.
- The stakeholders and their information needs.

- The required business and reporting capabilities.
- The key performance indicators required for monitoring business performance.
- The expected business outcomes and success criteria.

2. Business Background

Olist is a Brazilian e-commerce marketplace that connects customers with multiple sellers across different regions.

The available dataset contains approximately 100,000 e-commerce orders from 2016 to 2018 and includes information related to customers, orders, products, sellers, payments, reviews, and delivery.

The project aims to transform this transactional data into a centralized analytical reporting solution that enables stakeholders to understand business performance and identify opportunities for improvement.

3. Business Problem

Business stakeholders require a centralized reporting solution to understand and monitor key areas of e-commerce performance.

Without a centralized analytical solution, answering important business questions related to revenue, customers, products, sellers, payments, customer satisfaction, and delivery performance can be time-consuming.

The business needs the ability to answer questions such as:

- How much revenue is being generated?
- Which product categories generate the highest revenue?
- Which states and cities contribute the most revenue?
- Which payment methods are most preferred?
- How satisfied are customers?
- Are orders being delivered on time?
- Which sellers contribute the most revenue?
- How can customer retention and repeat purchases be improved?

4. Business Objectives

The proposed analytics solution should help the business:

1. Monitor overall sales and revenue performance.
2. Identify high-performing product categories.

3. Understand customer purchasing behavior.
4. Identify valuable customers and repeat-purchase opportunities.
5. Evaluate seller performance.
6. Analyze payment-method preferences.
7. Monitor customer satisfaction through review scores.
8. Measure delivery and logistics performance.
9. Identify regional sales opportunities.
10. Support data-driven business decision-making through interactive dashboards.

5. Project Scope

5.1 In Scope

The project includes:

- Customer analysis.
- Order analysis.
- Sales and revenue analysis.
- Product and category analysis.
- Seller performance analysis.
- Payment analysis.
- Customer review analysis.
- Delivery performance analysis.
- Geographic analysis.
- KPI reporting.
- SQL-based data analysis.
- Interactive Power BI dashboards.
- Business insights and recommendations.

5.2 Out of Scope

The following areas are outside the scope of the current project:

- Real-time transaction processing.
- Actual order management.
- Inventory management system implementation.
- Payment gateway implementation.

- Seller onboarding system.
- Customer-facing e-commerce application development.
- Machine-learning-based forecasting implementation.

6. Stakeholders

Stakeholder	Information Need
Executive / Management	Overall business performance and strategic KPIs
Sales Manager	Revenue trends, sales performance, categories, and regions
Customer/Marketing Team	Customer behavior, retention, reviews, and purchasing patterns
Product Manager	Product and category performance
Operations / Logistics Team	Delivery and shipping performance
Seller Management Team	Seller contribution and performance
Business Analyst	Requirements, insights, KPIs, and business recommendations
Data/BI Analyst	SQL analysis, data model, DAX, and dashboard development

7. Current State

The available e-commerce data is distributed across multiple related tables covering customers, orders, products, sellers, payments, reviews, and delivery information.

Analyzing these areas individually can make it difficult to obtain a consolidated view of business performance.

Stakeholders therefore require a structured analytical solution that brings relevant information together and presents it through understandable business dashboards.

8. Proposed Solution

The proposed solution is an end-to-end SQL and Power BI analytics solution.

Solution Flow

Raw Olist Dataset



SQL Data Exploration & Analysis



Data Modeling & Transformation



DAX KPI Creation



Power BI Interactive Dashboards



Business Insights



Business Recommendations

The Power BI solution consists of five dashboards:

1. Executive Dashboard
2. Sales Dashboard
3. Customer Dashboard
4. Product Dashboard
5. Delivery Dashboard

The dashboards allow stakeholders to analyze business performance using interactive Year and Quarter filters.

9. Business Requirements

BR-01 — Overall Performance Reporting

The solution should provide stakeholders with a consolidated view of overall business performance.

BR-02 — Sales Performance Analysis

The solution should enable stakeholders to monitor revenue, order performance, average order value, and sales trends.

BR-03 — Customer Analysis

The solution should provide visibility into customer volume, customer behavior, repeat purchases, and customer satisfaction.

BR-04 — Product Performance Analysis

The solution should enable stakeholders to identify top-performing product categories and evaluate product-level performance.

BR-05 — Seller Performance Analysis

The solution should provide visibility into seller contribution and identify high- and low-performing sellers.

BR-06 — Payment Analysis

The solution should show the distribution and usage of different payment methods.

BR-07 — Delivery Performance Analysis

The solution should enable stakeholders to monitor delivery time, shipping time, on-time delivery, and delayed deliveries.

BR-08 — Geographic Analysis

The solution should allow stakeholders to analyze revenue and customer distribution across states and cities.

BR-09 — Customer Satisfaction Analysis

The solution should provide review-score analysis to monitor customer satisfaction.

BR-10 — Interactive Reporting

The solution should allow stakeholders to filter and explore dashboard information dynamically.

10. Functional Requirements

ID	Functional Requirement
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FR-01	The dashboard shall display key business KPIs.
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FR-02	The dashboard shall display revenue trends over time.
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FR-03	Users shall be able to filter analysis by Year.
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FR-04	Users shall be able to filter analysis by Quarter.
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FR-05	Users shall be able to analyze sales by product category.
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FR-06	Users shall be able to analyze revenue by geographic region.
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FR-07	Users shall be able to analyze customer purchasing behavior.
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FR-08	Users shall be able to analyze seller performance.
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FR-09	Users shall be able to analyze payment-method preferences.
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FR-10	Users shall be able to analyze customer review scores.
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FR-11	Users shall be able to monitor delivery performance.
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FR-12	Dashboard visuals shall interact with applicable filters and selections.
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11. Non-Functional Requirements

ID Requirement

NFR-01 Dashboards should be easy for business users to understand.

NFR-02 KPI definitions should be consistent across dashboards.

NFR-03 Data displayed in dashboards should be based on validated analytical calculations.

NFR-04 Dashboard navigation should be intuitive.

NFR-05 Reports should provide clear and readable visualizations.

NFR-06 The solution should support interactive filtering and analysis.

12. KPI Requirements

The solution should provide key metrics including:

- Total Revenue
- Total Orders
- Unique Customers
- Average Order Value
- Gross Sales
- Profit / Estimated Profit
- Revenue per Customer
- Payment Transactions
- Average Delivery Time
- Shipping Time
- On-Time Delivery Rate
- Delayed Deliveries
- Average Review Score
- Revenue by Product Category
- Revenue by State
- Seller Revenue

Detailed KPI definitions and formulas will be maintained in the separate **KPI Documentation**.

13. Reporting Requirements

The reporting solution should provide five primary dashboards.

Executive Dashboard

Provides a high-level overview of overall business performance and major KPIs.

Sales Dashboard

Provides sales trends, revenue performance, average order value, category performance, and geographic sales analysis.

Customer Dashboard

Provides customer growth, purchasing behavior, repeat-customer analysis, and customer satisfaction insights.

Product Dashboard

Provides product and category performance, revenue contribution, freight analysis, and review performance.

Delivery Dashboard

Provides delivery time, shipping time, on-time delivery, delayed orders, and regional logistics analysis.

14. Assumptions

- The Olist dataset is representative of the business scenario being analyzed.
- The available historical dataset is sufficient for the defined analysis.
- Stakeholders will use the dashboards for analytical and decision-support purposes.
- KPI calculations are based on the available dataset fields and defined business logic.
- The current solution is intended for analytical reporting rather than operational transaction processing.

15. Constraints

- The dataset represents historical data from 2016–2018.
- The solution does not provide real-time reporting.
- Inventory and cost information is limited in the available dataset.
- Forecasting and machine-learning models are not included in the current implementation.
- Business results are dependent on the quality and completeness of the available dataset.

16. Success Criteria

The project will be considered successful if it:

1. Provides a centralized view of key e-commerce performance metrics.

2. Enables stakeholders to analyze sales, customers, products, sellers, and delivery performance.
3. Provides consistent and understandable KPIs.
4. Enables interactive analysis through Power BI dashboards.
5. Reduces the effort required to manually analyze business performance.
6. Generates actionable business insights and recommendations.
7. Supports data-driven decision-making across relevant business functions.

17. Approval / Sign-Off

Role	Name	Status	Date
Project Owner / Business Analyst	Sejal Sayam	Pending	—
Business Stakeholder	—	Pending	—
BI / Reporting Stakeholder	—	Pending	—

End of Document